

1) Why fiat depends on trust

Fiat money has little or no intrinsic value and is usually not backed by a valuable commodity like gold or silver. Its value depends mainly on people trusting the government, central bank, and overall economic system. As long as people believe the currency will continue to be accepted and managed properly, fiat works. But if trust weakens badly, people lose confidence in holding that money, and its value can collapse.

2) Hyperinflation of Germany 1923

Germany's hyperinflation after World War I is a classic example of what can happen when money printing grows out of control without real economic backing. Germany had war debt, reparations, industrial disruption, and low productivity, yet kept printing more and more money. This caused the value of the German mark to collapse so severely that by November 1923, one US dollar was worth over 4.2 trillion German marks. The lesson is that money can become nearly worthless if trust and monetary discipline break down.

3) Digital currency requirements

A good digital currency should prevent stealing, prevent double-spending, allow fair creation or minting, support peer-to-peer payments, avoid dependence on a third party always being online, avoid unnecessary intermediaries, and preserve privacy at a level similar to physical cash. These requirements are important because earlier digital payment systems usually solved only some of these problems, not all of them together. Bitcoin became important because it tried to satisfy these requirements in one decentralized design.

4) Traditional ePayments pros and cons

Traditional ePayments are useful because they are digital and allow people in different locations to pay each other conveniently. However, they are not truly peer-to-peer because banks or payment companies remain in the middle. These intermediaries can fail, delay payments, censor users, and observe full transaction details such as who paid whom, how much, and when. So traditional ePayments improve convenience but still keep central control and weak privacy.

5) Why banks being “in the way” matters

Banks being “in the way” matters because they become a central dependency for every transaction. If the bank fails, goes offline, freezes funds, delays payments, or refuses service, users lose control over their own ability to transact. Banks also gain full visibility into payment behavior, which reduces privacy. Blockchain and cryptocurrencies became attractive partly because they try to reduce this dependence on a trusted central intermediary.

6) Simple e-cash and double-spending

Simple e-cash tries to create digital peer-to-peer money, but it faces the core problem that a digital coin is just data and can be copied like a file. That means Alice could spend the same digital coin with Bob and then try to spend the same copied coin again with Charlie. This is called double-spending. So simple e-cash shows that making money digital is easy, but making it safely spendable only once is the real challenge.

7) Serial numbers in simple e-cash

Serial numbers are added in simple e-cash to make each digital coin unique. If the same serial number appears in two different spending attempts, the system can detect that the same coin has been used more than once. This helps identify double-spending, but it does not completely solve the problem in a fully peer-to-peer system. To prevent double-spending before it causes harm, a bank or central checker usually still needs to be online.

8) Privacy problem in simple e-cash

Simple e-cash has a privacy problem because the bank can see the serial number of the coin and connect it to the transaction history. Once the bank knows the serial number, it may be able to tell who withdrew the coin, who spent it, and possibly who received it. So even though the payment is digital, the bank still gains too much knowledge about the users’ transaction behavior. That means privacy is much weaker than with physical cash.

9) Chaumian e-cash and blinded signatures

Chaumian e-cash improves privacy by using blinded signatures. Alice chooses a serial number for a coin, blinds it, and sends the blinded version to the bank. The bank signs it without seeing the real serial number, and Alice later removes the blinding to obtain a valid bank signature on the real coin. This means the bank authorizes the coin but cannot easily link the withdrawal to the later spending, which gives privacy close to physical cash.

10) Why Chaumian e-cash still failed

Chaumian e-cash was technically strong on privacy, but it still failed commercially because it remained dependent on a trusted bank, could still be censored, and required online verification that was inconvenient in the 1990s. Governments disliked it because it was considered too private and potentially useful for tax evasion or money laundering. Banks also had little incentive to support a system that reduced their control over payments. So it was too private for governments, too inconvenient for users, and too unattractive for banks.

11) Desirable features of money

Good money should be durable, portable, divisible, fungible, widely acceptable, reasonably stable in value, limited in supply, and hard to counterfeit. These properties matter because money must work reliably for storage, exchange, and accounting. Whether money is physical or digital, it should still be easy to transfer, easy to divide, trusted by users, and resistant to forgery or uncontrolled inflation. These features are the standard checklist for evaluating any money system.

12) What Bitcoin solved from earlier digital currencies

Bitcoin addressed several weaknesses of earlier digital currencies by combining digital signatures for secure ownership, a global shared ledger to prevent double-spending, and Proof of Work to support fair decentralized block creation. Earlier systems either depended too much on a central bank, failed on privacy, or failed on double-spending prevention. Bitcoin's main contribution was showing that digital money could work in a decentralized way without requiring one always-online trusted intermediary. That is why it is seen as a breakthrough rather than just another digital token scheme.

13) Consensus in blockchains

Consensus in blockchains means reaching agreement among distributed participants about which transactions and blocks should become part of the shared ledger. This agreement must happen dynamically because new transactions keep arriving all the time. Consensus is needed so that conflicting transactions, like two attempts to spend the same money, do not both become accepted in the final ledger. Without consensus, different nodes could keep different incompatible histories and the system would stop being reliable.

14) Weighted majority

Weighted majority means that consensus is not based on simply counting how many nodes exist, because one dishonest user could create many fake nodes. Instead, each node is given influence based on some meaningful resource, such as hashing power in Proof of Work or stake in Proof of Stake. This makes attacks more expensive because power in the system is tied to costly commitment rather than cheap identity creation. Weighted majority is therefore a defense against fake-node domination.

15) PoW vs PoS in Week 4 framing

In Week 4, Proof of Work is presented as a system where nodes compete to solve a computationally expensive puzzle, and the first solver wins the right to add the block. This makes abuse costly through energy and computation. Proof of Stake is presented as an alternative that avoids heavy computation by requiring validators to lock up cryptocurrency, with influence depending on stake and cheating punished by loss of deposit. So PoW ties consensus power to work, while PoS ties it to staked value.

16) Smart contracts

A smart contract is a program that runs on the blockchain and automatically executes an agreement when predefined conditions are met. Instead of needing a middleman to manually enforce terms, the contract code handles the logic itself. For example, a crop insurance smart contract can automatically release payment if rainfall falls below a specified threshold. Smart contracts are important because they make blockchains useful for more than just currency transfer.

17) Oracle / external data

An oracle is a mechanism that brings real-world data into a blockchain system so that smart contracts can react to outside events. Since a blockchain cannot directly know things like weather, market prices, or sports results on its own, it depends on oracles to provide that information. In the crop insurance example, rainfall data must come from a source outside the blockchain. So oracles act as bridges between on-chain logic and off-chain reality.

18) Tokenization

Tokenization means representing an asset as a digital token on a blockchain or similar system. The asset could be money, stocks, real estate, gold, or something else of value. The point of tokenization is that once the asset is represented as a token, it may become easier to transfer, divide, trade peer-to-peer, or use in programmable financial systems. Tokenization is important because it expands blockchain use beyond cryptocurrencies into broader asset representation.

19) Layer 2

Layer 2 refers to a secondary protocol built on top of an existing blockchain to improve speed, efficiency, or functionality. Many main blockchains are secure but slow or expensive, so Layer 2 systems try to handle more transactions or offer extra features while still relying on the main chain beneath them. Examples include Polygon on Ethereum and Stacks for Bitcoin. The main idea is to scale or extend blockchain systems without replacing the base layer entirely.

20) CBDC vs digital fiat

CBDC stands for Central Bank Digital Currency and is digital money issued or governed directly by the central bank. Digital fiat in everyday banking, on the other hand, usually exists as balances managed by commercial banks rather than by the central bank directly. So although both may be digital, the governance structure is different. CBDCs give the state more direct control and visibility, while ordinary digital bank money still depends on the commercial banking layer.